

Object Literals

used to store keyed collections & complex entities.

property $\Rightarrow$ (key, value) pair

↳ Collections of properties $\rightarrow$ Objects

Ex:

(Object)
student

name	"Shradha"	Property 1
age	23	Property 2
marks	94.4	Property 3

↓ key ↓ value

Get values

```
let student = {  
  name : " ABC ",  
  marks : 94.4  
};
```

→

obj ["key"]
obj. key

Ex: student.name

Conversion in Get values

- JS automatically converts objects keys to strings
- Even if we made the no. as a key, then the no. will be converted to string.

Add / Update values

```
Ex:  const student = {  
      name : "Aman",  
      age : 23 ,  
      marks : 94.4 ,  
      city : "Delhi"  
    } ;
```

- change city to "Mumbai"

```
student.city = "Mumbai" ;
```

o/p : 'Mumbai'

- Add a new property ; gender : "Female"

```
student.gender = "Female" ;
```

o/p : 'Female'

- Delete marks property

```
delete student.marks ;
```

o/p : true

Object of Objects (nesting)

- Storing info of multiple students

```
const class Info = {  
  aman : {  
    grade : " A+ ",  
    city : " Delhi "  
  },  
  shradha : {  
    grade : " A ",  
    city : " Pune "  
  },  
  karan : {  
    grade : " O ",  
    city : " Mumbai "  
  }  
};
```

object

object

object

Get values :

→ `classInfo.aman`

o/p : `{ grade : 'A+', city : 'Delhi' }`

→ `classInfo.shradha.city`

o/p : `'Pune'`

→ `classInfo.shradha.city = "Gurgaon";`

o/p : `'Gurgaon'`

Array of Objects

- storing info of multiple students

```
const classInfo = [  
  {  
    name : "aman",  
    grade : "A+",  
    city : "Delhi"  
  },  
  {  
    name : "shradha",  
    grade : "A",  
    city : "Pune"  
  },  
  {  
    name : "karan",  
    grade : "O",  
    city : "mumbai"  
  }  
]
```

];

Get values : `classInfo (0).name`

o/p : aman

`classInfo (0).name = "Pradeep";`

o/p : Pradeep

Math Objects

Ex: `Math.PI`

`Math.E`

Methods : `Math.abs(n)`

`Math.pow(a, b)`

`Math.floor(n)`

`Math.ceil(n)`

`Math.random()` → generates random no.

0 → 1 (excluding 1)

Q) Generate random no. b/w 1 to 10

1) `let num = Math.random();` // 0.4674774....

2) `num = num * 10;` // 4.674774....

3) `num = Math.floor(num);` // 4

4) `num = num + 1` // 5

`Math.floor(Math.random() * 10) + 1;`

